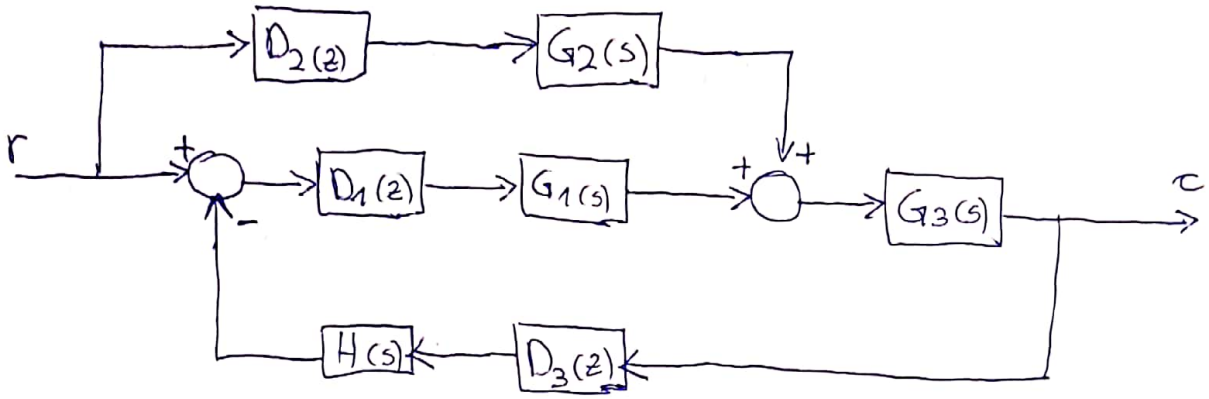
18.04.2016. K1

① Дигитални систем управљања дат је на слици.

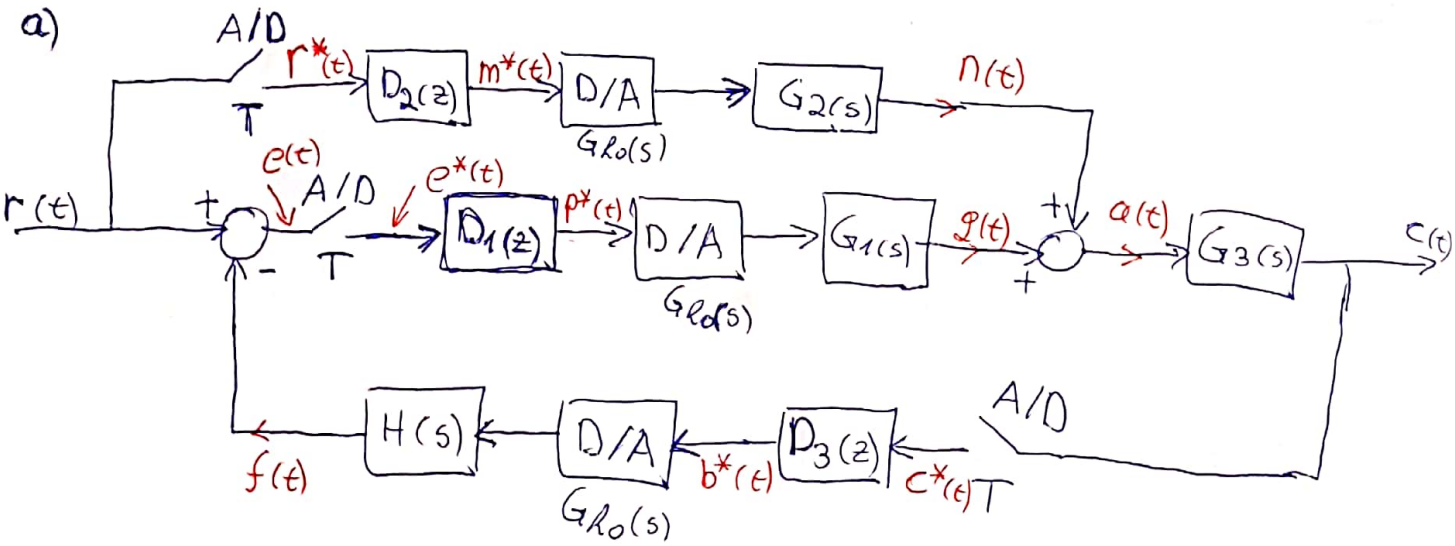


a) Одредити мјесто(а) гдје је потребно поставити одабирач(е) и како(а) задржке нултог реда.

b) Одредити ф-ју дискретног преноса $\frac{C(z)}{R(z)}$.

РЈЕШЕЊЕ:

a)



b) $C(s) = G_3(s) \cdot A(s) = G_3(s)\left[N(s) + Q(s)\right]$

$N(s) = G_{h0}(s) \cdot G_2(s) \cdot M^*(s) = G_{h0}(s) \cdot G_2(s) \cdot D_2(z) \cdot R^*(s)$

$Q(s) = G_{h0}(s) \cdot G_1(s) \cdot P^*(s) = G_{h0}(s) \cdot G_1(s) \cdot D_1(z) \cdot E^*(s)$

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$$C(s) = G_{\ell o}(s) \cdot G_2(s) \cdot G_3(s) \cdot D_2(z) \cdot R^*(s) + G_{\ell o}(s) \cdot G_1(s) \cdot G_3(s) \cdot D_1(z) \cdot E^*(s)$$

$$E(s) = R(s) - F(s) = R(s) - G_{\ell o}(s) \cdot H(s) \cdot B^*(s)$$

$$E(s) = R(s) - G_{\ell o}(s) \cdot H(s) \cdot D_3(z) \cdot C^*(s) \xrightarrow[\text{ТАФ,ДП}]{}$$

$$E^*(s) = R^*(s) - G_{\ell o} H^*(s) \cdot D_3(z) \cdot C^*(s) \quad / Z$$

$$\cancel{C(s) = G_{\ell o}(s) \cdot G_2(s) \cdot G_3(s) \cdot D_2(z) \cdot R^*(s) +}$$

$$\cancel{+ G_{\ell o}(s) \cdot G_1(s) \cdot G_3(s) \cdot D_1(z) [R^*(s)}$$

$$E(z) = R(z) - G_{\ell o} H(z) \cdot D_3(z) \cdot C(z)$$

$$C(s) = G_{\ell o}(s) \cdot G_2(s) \cdot G_3(s) \cdot D_2(z) \cdot R^*(s) + G_{\ell o}(s) \cdot G_1(s) \cdot G_3(s) \cdot D_1(z) \cdot E^*(s)$$

$$\xrightarrow[\text{ТАФ,ДП}]{} C^*(s) = G_{\ell o} G_2 G_3^*(s) \cdot D_2(z) \cdot R^*(s) + G_{\ell o} G_1 G_3^*(s) \cdot D_1(z) \cdot E^*(s) \quad / Z$$

$$C(z) = G_{\ell o} G_2 G_3(z) \cdot D_2(z) \cdot R(z) + G_{\ell o} G_1 G_3(z) \cdot D_1(z) \cdot E(z)$$

$$C(z) = G_{\ell o} G_2 G_3(z) \cdot D_2(z) \cdot R(z) + G_{\ell o} G_1 G_3(z) \cdot D_1(z) [R(z) - G_{\ell o} H(z) \cdot D_3(z) \cdot C(z)]$$

$$C(z) [1 + G_{\ell o} G_1 G_3(z) \cdot D_1(z) \cdot G_{\ell o} H(z) \cdot D_3(z)] =$$

$$= R(z) [G_{\ell o} G_2 G_3(z) \cdot D_2(z) + G_{\ell o} G_1 G_3(z) \cdot D_1(z)]$$

$$\boxed{\frac{C(z)}{R(z)} = \frac{G_{\ell o} G_2 G_3(z) \cdot D_2(z) + G_{\ell o} G_1 G_3(z) \cdot D_1(z)}{1 + G_{\ell o} G_1 G_3(z) \cdot D_1(z) \cdot G_{\ell o} H(z) \cdot D_3(z)}}$$

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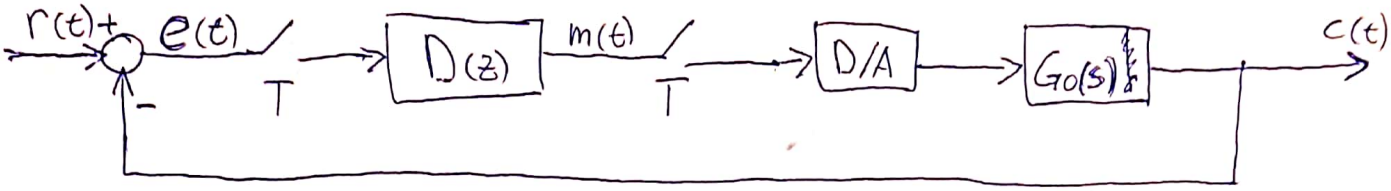
(2.) Импулсни систем управљања дат је на слици.



a) Одредити ф-ју дискретног преноса ако је регулатор представљен са $D(z) = \frac{z}{0.5z - 0.4}$, а објекат управљања са $G_0(s) = \frac{2}{s+2}$. Периода одмјеравања је $T = 1[s]$.

b) Одредити вриједности сигнала $c(t)$ у тренутку $t = 1.25T$ и $t = 2.25T$ ако се на улаз доведе јединични одскочни сигнал.

РЈЕШЕЊЕ:

a) ~~$C(s) = G_{h0}(s) \cdot G_0(s) \cdot M^*(s)$; $M^*(s) = D(z)$~~

$$\left.\begin{aligned} C(s) &= G_{h0}(s) \cdot G_0(s) \cdot M^*(s) \\ M^*(s) &= D(z) \cdot E^*(s) \end{aligned}\right\} \Rightarrow C(s) = G_{h0}(s) \cdot G_0(s) \cdot D(z) \cdot E^*(s)$$

$$\underset{\text{ТАФДП}}{\Longrightarrow} \quad C^*(s) = G_{h0} G_0^*(s) \cdot D(z) \cdot E^*(s) \;/ z$$

$$C(z) = G_{h0} G_0(z) \cdot D(z) \cdot E(z)$$

$$E(s) = R(s) - C(s) \underset{\text{ТАФДП}}{\Longrightarrow} E^*(s) = R^*(s) - C^*(s) \;/ z$$

$$E(z) = R(z) - C(z)$$

$$C(z) = G_{h0} G_0(z) \cdot D(z) \cdot [R(z) - C(z)]$$

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$$C(z)\left[1 + G_{h_0}G_0(z)\cdot D(z)\right] = G_{h_0}G_0(z)\cdot D(z)\cdot R(z)$$

$$\boxed{\frac{C(z)}{R(z)} = \frac{G_{h_0}G_0(z)\cdot D(z)}{1+G_{h_0}G_0(z)\cdot D(z)}}$$

$$G_{h_0}G_0(z) = Z\left\{\frac{1-e^{-sT}}{s}\cdot\frac{2}{s+2}\right\} = (1-z^{-1})\,Z\left\{\frac{2}{s(s+2)}\right\}$$

$$\frac{2}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} = \frac{As+2A+Bs}{s(s+2)} \Rightarrow \left.\begin{matrix} A+B=0 \\ 2A=2 \end{matrix}\right\} \Rightarrow \begin{matrix} A=1 \\ B=-1 \end{matrix}$$

$$G_{h_0}G_0(z) = (1-z^{-1})\,Z\left\{\frac{1}{s}-\frac{1}{s+2}\right\} = \frac{z-1}{z}\cdot\left[\frac{z}{z-1}-\frac{z}{z-e^{-2T}}\right]$$

$T = 1[s]$

$$G_{h_0}G_0(z) = \frac{z-1}{z}\left[\frac{z}{z-1}-\frac{z}{z-e^{-2}}\right] = 1 - \frac{z-1}{z-0.1353} =$$

$$= \frac{\cancel{z}-0.1353-\cancel{z}+1}{z-0.1353} \Rightarrow G_{h_0}G_0(z) = \frac{0.8647}{z-0.1353}$$

$$\frac{C(z)}{R(z)} = \frac{\dfrac{0.8647}{(z-0.1353)}\cdot\dfrac{z}{(0.5z-0.4)}}{1+\dfrac{0.8647}{(z-0.1353)}\cdot\dfrac{z}{(0.5z-0.4)}}$$

$$\frac{C(z)}{R(z)} = \frac{\dfrac{0.8647z}{\cancel{(z-0.1353)(0.5z-0.4)}}}{\dfrac{(z-0.1353)(0.5z-0.4)+0.8647z}{\cancel{(z-0.1353)(0.5z-0.4)}}} = \frac{0.8647z}{0.5z^2-0.46765z+0.05412+0.8647z}$$

$$\frac{C(z)}{R(z)} = \frac{0.8647z}{0.5z^2+0.39705z+0.05412}$$

$\cancel{\frac{C(z)}{R(z)}}$

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$$0.5z^2 + 0.39705z + 0.05412 = 0$$

$$z_{1,2} = \frac{-0.39705 \pm \sqrt{0.39705^2 - 4 \cdot 0.5 \cdot 0.05412}}{2 \cdot 0.5}$$

$$z_{1,2} = \frac{-0.39705 \pm 0.2223}{1} \Rightarrow \begin{array}{l} z_1 = -0.17475 \\ z_2 = -0.61935 \end{array}$$

$$\boxed{\frac{C(z)}{R(z)} = \frac{0.8647z}{0.5(z+0.17475)(z+0.61935)}}$$

b) $$C^*(s) = \frac{G_{h_0} G_0^*(s) \cdot D(z) \cdot R^*(s)}{1 + G_{h_0} G_0^*(s) \cdot D(z)} \Big/ \cdot \left(e^{-(1-m)sT}\right)$$

$$\left[C(s) e^{-(1-m)sT}\right]^* = \frac{\left[G_{h_0} G_0(s) e^{-(1-m)sT}\right]^* \cdot D(z) \cdot R^*(s)}{1 + G_{h_0} G_0^*(s) \cdot D(z)} \Big/ z$$

$$C(z,m) = \frac{G_{h_0} G_0(z,m) \cdot D(z)}{1 + G_{h_0} G_0(z) \cdot D(z)} \cdot R(z)$$

$$G_{h_0} G_0(z,m) = Z_m \left\{ \frac{1 - e^{-sT}}{s} \cdot \frac{2}{s+2} \right\} = (1 - z^{-1}) Z_m \left\{ \frac{2}{s(s+2)} \right\} =$$

$$= (1 - z^{-1}) Z_m \left\{ \frac{1}{s} - \frac{1}{s+2} \right\} = \frac{z-1}{z} \left[\frac{1}{z-1} - \frac{e^{-2mT}}{z - e^{-2T}} \right]$$

$T = 1\,[s]$

$$G_{h_0} G_0(z,m) = \frac{z-1}{z} \left[\frac{1}{z-1} - \frac{e^{-2m}}{z - e^{-2}} \right] =$$

$$= \frac{1}{z} - \frac{(z-1)e^{-2m}}{z(z - 0.1353)}$$

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$c[kT-(1-m)T]$ $\quad C[1.25T],\ C[2.25T] = ?$

$m = 0.25 \quad , \lambda = 1 - m = 0.75$

$$G_{h_0}G_0(z,m) = G_{h_0}G_0(z, 0.25) = \frac{1}{z} - \frac{e^{-2\cdot 0.25}(z-1)}{z(z-0.135^3)} =$$

$$= \frac{z - 0.1353 - 0.6065z + 0.6065}{z(z-0.1353)}$$

$$G_{h_0}G_0(z,m) = \frac{0.3935z + 0.4712}{z(z-0.1353)}$$

$$C(z,m) = \frac{\frac{0.3935z+0.4712}{\cancel{z}(z-0.1353)} \cdot \frac{\cancel{z}}{(0.5z-0.4)}}{1 + \frac{0.8647}{(z-0.1353)} \cdot \frac{z}{(0.5z-0.4)}} \cdot \frac{z}{z-1}$$

улаз јединични одскочни сигнал

$$= \frac{\frac{0.3935z+0.4712}{(z-0.1353)\cancel{(0.5z-0.4)}}}{\frac{(z-0.1353)(0.5z-0.4)+0.8647z}{\cancel{(z-0.1353)(0.5z-0.4)}}} \cdot \frac{z}{z-1}$$

$$= \frac{0.3935z+0.4712}{0.5z^2-0.46765z+0.05412+0.8647z} \cdot \frac{z}{z-1} =$$

$$= \frac{0.3935z+0.4712}{0.5z^2+0.39705z+0.05412} \cdot \frac{z}{z-1} =$$

$$= \frac{0.3935z^2+0.4712z}{0.5z^3-0.5z^2+0.39705z^2-0.39705z+0.05412z-0.05412}$$

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$$G(z,m) = \frac{0.3935 z^2 + 0.4712 z}{0.5 z^3 - 0.10295 z^2 - 0.34293 z - 0.05412}$$

$$c[kT-(1-m)T] = Z_m^{-1}\{G(z,m)\}$$

$$(0.3935 z^2 + 0.4712 z) : (0.5 z^3 - 0.10295 z^2 - 0.34293 z - 0.05412) = 0.787 z^{-1} + 1.10444 z^{-2}$$

$$-(0.3935 z^2 - 0.081022 z - 0.2699 - 0.0426 z^{-1})$$

$$= 0.55222 z + 0.2699 + 0.0426 z^{-1}$$

$$\boxed{\begin{aligned} c[1.25T] &= 0.787 \\ c[2.25T] &= 1.10444 \end{aligned}}$$

3.) Континуалан систем представљен је моделом у простору стања

$$\dot{X}(t) = \begin{bmatrix} -3 & 0 \\ 4 & -2 \end{bmatrix} X(t) + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u(t)$$

$$Y(t) = \begin{bmatrix} -2 & 1 \end{bmatrix} X(t).$$

a) Одредити дискретан модел система у простору стања.

b) Одредити Ф-ју дискретног преноса система.

Усвојити да је $X(0)=0$ и период одмјеравања $T=1[s]$.

РЈЕШЕЊЕ:

a) $X(k+1) = E\,X(k) + F u(k)$

$Y(k) = C\,X(k)$

$E = e^{AT}$, $F = \int_0^T e^{AT} B\,dt$

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$$E = e^{AT} = \mathcal{L}^{-1}\left\{(sI-A)^{-1}\right\} = \mathcal{L}^{-1}\left\{\left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -3 & 0 \\ 4 & -2 \end{bmatrix}\right)^{-1}\right\}\Bigg|_{t=T} =$$

$$= \mathcal{L}^{-1}\left\{\begin{bmatrix} s+3 & 0 \\ -4 & s+2 \end{bmatrix}^{-1}\right\}\Bigg|_{t=T} = \mathcal{L}^{-1}\left\{\frac{1}{(s+3)(s+2)}\begin{bmatrix} s+2 & 0 \\ 4 & s+3 \end{bmatrix}\right\}\Bigg|_{t=T} =$$

$$= \mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{1}{s+3} & 0 \\ \frac{4}{(s+3)(s+2)} & \frac{1}{s+2} \end{bmatrix}\right\}\Bigg|_{t=T}$$

$$\frac{4}{(s+3)(s+2)} = \frac{A}{s+2} + \frac{B}{s+3} = \frac{As+3A+Bs+2B}{(s+2)(s+3)} \Rightarrow \begin{array}{l} A+B=0 \\ 3A+2B=4 \\ A=4,\ B=-4 \end{array}$$

$$E = \mathcal{L}^{-1}\left\{\begin{bmatrix} \frac{1}{s+3} & 0 \\ \frac{4}{s+2} - \frac{4}{s+3} & \frac{1}{s+2} \end{bmatrix}\right\}\Bigg|_{t=T} = \begin{bmatrix} e^{-3t} & 0 \\ 4(e^{-2t}-e^{-3t}) & e^{-2t} \end{bmatrix}\Bigg|_{t=T}$$

$$E(T) = \begin{bmatrix} e^{-3T} & 0 \\ 4(e^{-2T}-e^{-3T}) & e^{-2T} \end{bmatrix}, \quad T = 1\,[s]$$

$$E(T) = \begin{bmatrix} 0.0498 & 0 \\ 0.3422 & 0.1353 \end{bmatrix}$$

$$F(T) = \int_0^T e^{At} B\,dt = \int_0^T \begin{bmatrix} e^{-3t} & 0 \\ 4(e^{-2t}-e^{-3t}) & e^{-2t} \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} dt =$$

$$= \int_0^T \begin{bmatrix} e^{-3t} \\ 6e^{-2t} - 4e^{-3t} \end{bmatrix} dt = \begin{bmatrix} -\frac{1}{3}e^{-3t} \\ -3e^{-2t} + \frac{4}{3}e^{-3t} \end{bmatrix}\Bigg|_0^T =$$

$$= \begin{bmatrix} -\frac{1}{3}(e^{-3T}-1) \\ -3e^{-2T} + \frac{4}{3}e^{-3T} + 3 - \frac{4}{3} \end{bmatrix} = \begin{bmatrix} 0.3167 \\ 1.3270 \end{bmatrix}$$

$$\boxed{\begin{array}{l} x(k+1) = Ex(k) + Fu(k) = \begin{bmatrix} 0.0498 & 0 \\ 0.3422 & 0.1353 \end{bmatrix} x(k) + \begin{bmatrix} 0.3167 \\ 1.3270 \end{bmatrix} u(k) \\ y(k) = Cx(k) = \begin{bmatrix} -2 & 1 \end{bmatrix} x(k) \end{array}}$$

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b) $X(k+1) = E\,x(k) + F\,u(k) \;/\; Z$ | $Y(k) = C\,x(k) \;/\; Z$

$$z\,X(z) - z\underbrace{x(0)}_{0} = E\,X(z) + F\,U(z) \qquad Y(z) = C\,X(z)$$

$$(zI-E)^{-1} \Big/ (zI - E)X(z) = F\,U(z)$$

$$X(z) = (zI - E)^{-1} F\,U(z)$$

$$Y(z) = C\,X(z) = C\,(zI - E)^{-1} F\,U(z)$$

$$G(z) = \frac{Y(z)}{U(z)} = C\,(zI - E)^{-1} F$$

$$G(z) = \begin{bmatrix} -2 & 1 \end{bmatrix} \left(\begin{bmatrix} z & 0 \\ 0 & z \end{bmatrix} - \begin{bmatrix} 0.0498 & 0 \\ 0.3422 & 0.1353 \end{bmatrix} \right)^{-1} \begin{bmatrix} 0.3167 \\ 1.3270 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 1 \end{bmatrix} \begin{bmatrix} z - 0.0498 & 0 \\ -0.3422 & z - 0.1353 \end{bmatrix}^{-1} \begin{bmatrix} 0.3167 \\ 1.3270 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 1 \end{bmatrix} \cdot \frac{1}{(z-0.0498)(z-0.1353)} \begin{bmatrix} z-0.1353 & 0 \\ 0.3422 & z-0.0498 \end{bmatrix} \begin{bmatrix} 0.3167 \\ 1.3270 \end{bmatrix}$$

$$= \frac{1}{(z-0.0498)(z-0.1353)} \begin{bmatrix} -2 & 1 \end{bmatrix} \begin{bmatrix} 0.3167z - 0.0428 \\ 0.1084 + 1.3270z - 0.0661 \end{bmatrix}$$

$$= \frac{1}{(z-0.0498)(z-0.1353)} \begin{bmatrix} -2 & 1 \end{bmatrix} \begin{bmatrix} 0.3167z - 0.0428 \\ 1.3270z + 0.0423 \end{bmatrix} =$$

$$= \frac{-0.6334z + 0.0856 + 1.3270z + 0.0423}{(z-0.0498)(z-0.1353)}$$

$$\boxed{G(z) = \frac{0.6936z + 0.1279}{(z-0.0498)(z-0.1353)}}$$

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